

Section: **3.2 – MECHANICALLY FASTENED JOINTS**

Chapter: **3250**

Title: **LUGS AND PINS**

Revision: **1.0**

Date: **27/02/2023**

1 INTRODUCTION

1.1 OBJECT

The most typical configuration of the connections between two structural components, where one is hinged in the fixed part (access doors, cowlings, ...), consist of a line of fittings terminated with lugs and a pin-bolt joining both parts.

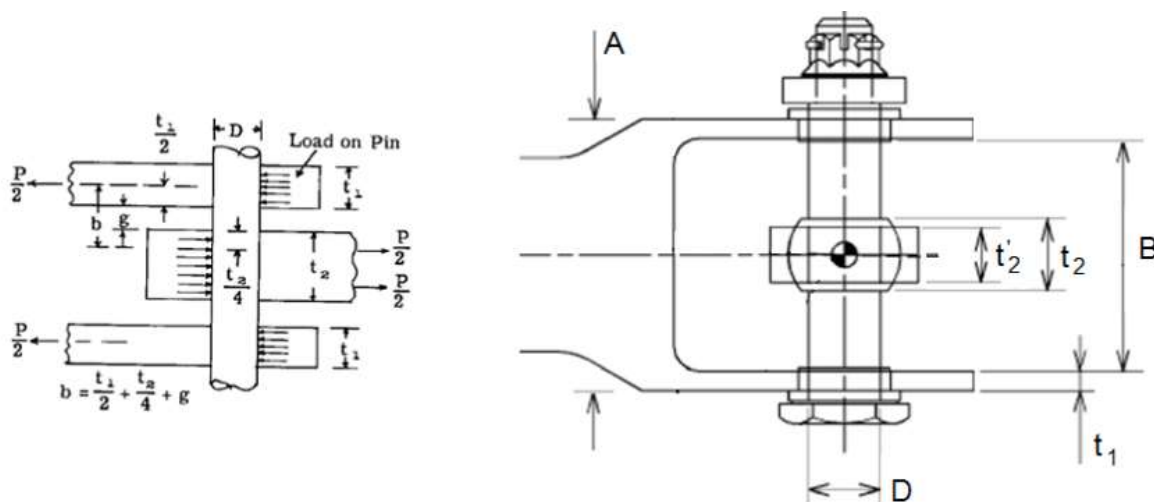


Figure 1. Bruhn's fig. D1.16 and typical pinned-lugs connection with ball-bearing in centre lug

1.2 SCOPE AND LIMITATIONS

Static strength analysis of metallic pinned lugs (clevis fastened joints).

2 METHOD DESCRIPTION

Bruhn's approach, in chapter D1.14 "Bolt bending strength", is used for calculating either the loads at the clevis-lugs (gap g zero at the critical side) and the bending moment in the pin-bolt (critical if the male lug is centred, i.e. gaps at both sides equal). For that same purpose, the equivalent methods in HSB 26101-02 "Static failure load of bolts in metallic double shear lugs" or ESDU91008 "STRENGTH OF LUGS" can be used.

However, exploitation of the load-peaking methodology proposed in those references shall be prevented during any of the sizing design phases (only permitted for the final certification/qualification of the part).

For calculating the strength of straight and tapered lugs, Ekvall's method (Ref. 4), the same described in the HSB 26101-02, is preferred. If the lug does not match that definition, but its section is reduced at the root side (necked lug) an alternative methodology is needed. See details in following paragraphs.

2.1 LUGS STRENGTH

The critical load for the single/male-lug is P .

For any of the clevis-lugs the critical load is obtained by stating simply the forces and moments equilibrium at some point. If P_1 and P_2 are the reaction forces at clevis lugs and b_1 and b_2 their distances to the male centre lug, using their centre planes as reference stations:

$$P = P_1 + P_2 \quad P_1 \cdot (b_1 + b_2) = P \cdot b_2 \quad \rightarrow \quad P_1 = P \cdot b_2 / (b_1 + b_2) \quad , \quad P_2 = P \cdot b_1 / (b_1 + b_2)$$

In case of floating male lug, the critical load for the others is obtained assuming the single/male-lug is in contact with the clevis-lug (i.e. the gap g is zero at the critical side). The effective lever arms for each of the clevis-lugs and the most critically loaded are then:

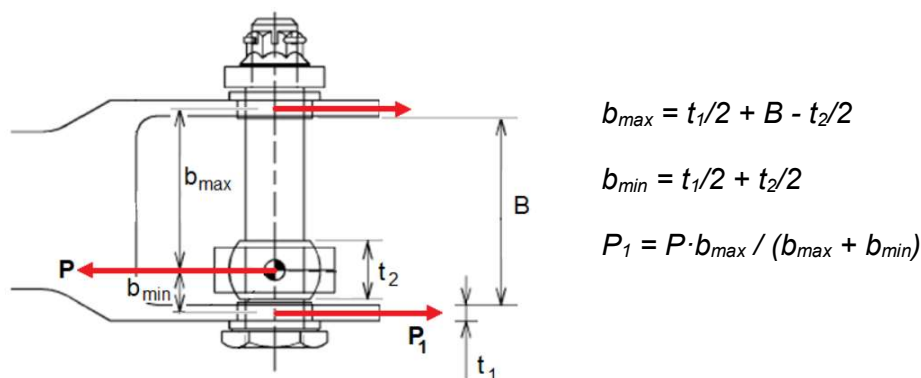


Figure 2. Calculation of critical load in clevis-lugs

2.1.1 Strength of straight and tapered lugs

The method proposed for calculating the in-plane strength of the lug is that proposed by the HSB 26101-01 (Ref. 2). This method is, on the other hand, mainly supported in J. C. Ekvall's work as described in his article "STRENGTH ANALYSIS OF PIN-LOADED LUGS" (1986).

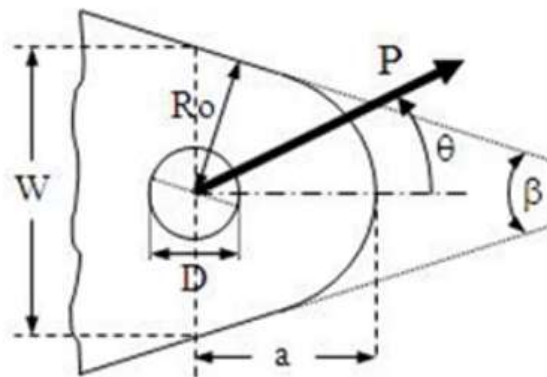


Figure 3. Tapered lug loaded within its plane

Some controversy aroused about the need of applying the statistical correction factors obtained by Ekvall (Ref. 4) in his regression analyses of lugs tests data. The HSB method definition includes a penalization factor, on top of the expression proposed to predict the lug strength, equal to 0.84. This knock down factor on the result of the regression curve corresponds to the 0.837 factor calculated by Ekvall to get the one side tolerance for 99% probability and 95% confidence (thus, the A-basis value). On the other hand, the material ultimate tensile strengths used by Ekvall in his correlation study were "the average of tensile tests conducted on specimens fabricated from the lug material". In other words, the variability observed in that correlation could be blamed on the natural dispersion on material properties and not really in any bias of the method itself. Then, if the stress engineer uses the right material design value for the ultimate tensile strength which corresponds to his case of use (B-basis or A-basis), it can be concluded that it would not be strictly necessary to apply additional penalization factors. Nevertheless, applying the extra penalization factor (e.g. the 0.84 proposed in the HSB) would be an option of programme stress margin policies (e.g. for critical joints, or for early stages of the project, ..).

2.1.2 Strength of necked lugs

Ekvall's method does not provide coverage to necked lugs, those with sections at the root span of width lower than the cross section at hole's centre (see following figure). For this kind of lugs the method described in the books of Bruhn, Niu or the ESDU91008, based in the proposal of Melcon and Hoblit (Ref. 10) will be used instead.

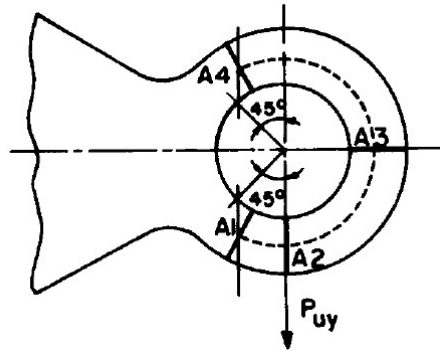


Figure 4. Example of necked lug and section areas applicable in strength analysis

A recommended design policy is to perform the strength analysis of the lugs (and bushings and pin if applies to the assembly configuration) considering an additional lateral out-of-plane load of 10% of the in-plane force P aligned with the pin direction, in order to cover potential, and probable, misalignments and eccentricities in real assembled parts.

2.2 PIN/BOLT BENDING

For floating male lugs, the critical bending moment in the pin-bolt is calculated with the male/single-lug centred in the clevis. Gap, critical lever arm and maximum bending moment are, then:

$$g = (B - t_2) / 2$$

$$b = t_2 / 4 + g + t_1 / 2 \quad (*) \text{ Note the lever arm here does not reach the mid-plane of the male lug}$$

$$M = P \cdot b / 2$$

In fixed lugs configurations, critical bending moment can be also at the male lug station with relationships shown previously.

No yield at Limit Load

The no-detrimental-permanent-deformation check of the bolt in limit loads can done using classical elastic beams theory. The corresponding Reserve Factor shall be the minimum between:

$$\text{Bending at outer fibres:} \quad RF_{y,b} = F_y = \text{Min}(F_{ty}, F_{cy}) / \sigma_{b,LL}$$

$$\text{Shear at centre:} \quad RF_{y,s} = F_{sy} / \tau_{LL} \approx (F_{ty} + F_{cy}) / (2 \cdot 3^{0.5}) / \tau_{LL}$$

In case the bolt is subjected, in addition, to a pretension force (P_t) and/or to an applied axial-tension force (T), the formulas for the failure criterion (no yield at Limit Load) and Reserve Factor must be slightly modified:

$$[P_t / A_b + RF_{y,bt} (T_{LL} / A_b \pm \sigma_{b,LL})] / F_y = 1 \quad \rightarrow \quad RF_{y,bt} = (F_y - P_t / A_b) / (T_{LL} / A_b \pm \sigma_{b,LL})$$

...in this case the yield allowable stress shall be F_{ty} or F_{cy} consistently with the sign of the stress at the point of analysis in axial plus bending loads.

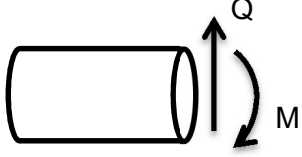
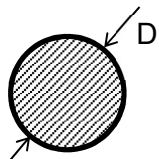
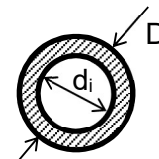
Useful formulas for linear-elastic analysis of circle sections 	 Solid circle section	 Hollow circle section
Area	$A_b = \pi \cdot (D/2)^2 = \pi \cdot D^2/4$	$A_b = \pi \cdot [(D/2)^2 - (d_i/2)^2] = \pi \cdot (D^2 - d_i^2)/4$
Bending inertia	$I_b = \pi \cdot (D/2)^4/4 = \pi \cdot D^4/64$	$I_b = \pi \cdot (D^4 - d_i^4)/64$
Bending stress at outer fibre: $\sigma_{b,LL} = M_{LL} \cdot (D/2) / I_b$	$\sigma_{b,LL} = 32 \cdot M_{LL} / (\pi \cdot D^3)$	$\sigma_{b,LL} = 32 \cdot M_{LL} / [\pi \cdot (D^4 - d_i^4)/D]$
Shear stress at centre $\tau_{LL} = K_s \cdot Q_{LL} / A_b$	$K_s = 4/3$	$K_s = (4/3) \cdot [(D^2 + d_i^2) / (D^2 - d_i^2)]$
Shape Factor (plastic bending) $SF = M_{plastic} / M_{yield}$	$SF = 1.698$	$SF = 1.698 \cdot (D^4 - d_i^3 \cdot D) / (D^4 - d_i^4)$

Table 1. Some useful formulas for circle sections (see Roark's book, TABLE A.1, for details)

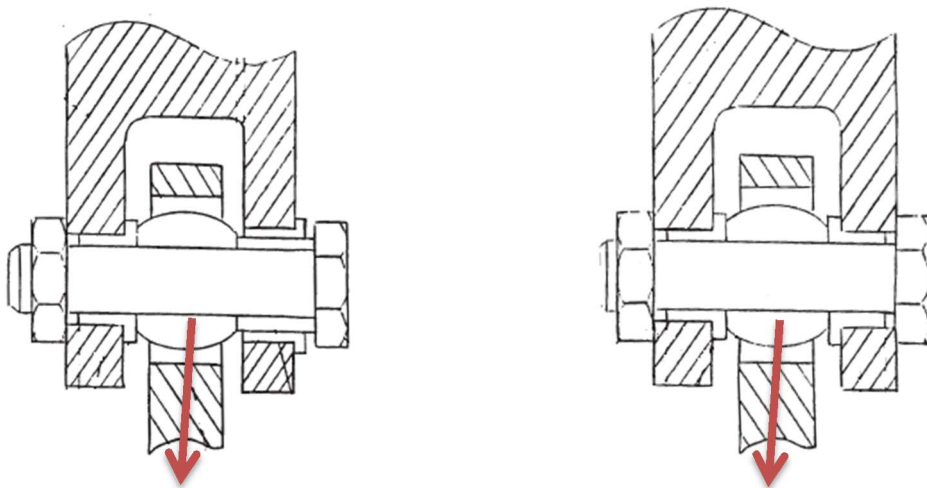


Figure 5. Pinned-lugs configurations for reaction of lateral load in one (left) or two (right) lugs

Critical shear load Q in the pin cross section shall be the maximum between P_1 and P_2 values, calculated as shown in previous pages, depending on the clevis assembly configuration.

No failure at Ultimate Load

The ultimate strength of the bolt and the static margin vs shear and bending can be calculated using a classical quadratic interaction:

$$RF = 1 / [(Q_{UL}/Q_{all})^2 + (M_{UL}/M_{all})^2]^{0.5}$$

Where M_{all} is the ultimate bending strength of the pin, accounting for material non-linearity, calculated using a specific software (e.g. FLEXA), Cozzone's approach (see Bruhn's C3.4) or the HSB 52130-01 method (see Ref. 6).

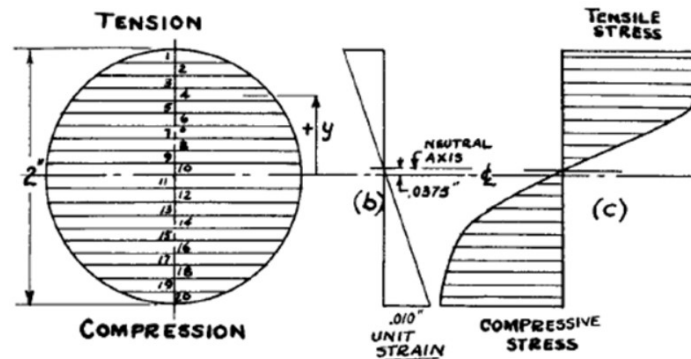


Figure 6. Bruhn's figure in chapter C3.3 Bending Strength of a Solid Round Bar

...and shear and tension ultimate allowable strengths of the pin-bolt are respectively:

$$Q_{all} = F_{su} \cdot A_b \quad (*)$$

$$T_{all} = F_{tu} \cdot A_b$$

(*) The HSB 26101-02 (Ref. 3) offers more sophisticated expressions for the shear ultimate strength in case of hollow circle sections

In case the bolt is pre-tensioned (typically vs a set of bushings fixing the position of the inner ball-bearing), or even sustaining an external applied load (with some of those type of assemblies possible if out-plane loads are considered) the bending allowable moment of the pin/bolt section shall be calculated taking into account the additional tension load. Reserve Factor can be obtained, using classical interaction formulas found in literature, and taking into account that pretension P_t is a constant parameter (not dependant of applied loads), as...

$$1 = (RF \cdot Q_{UL}/Q_{all})^2 + [P_t/T_{all} + RF \cdot (T_{UL}/T_{all} + M_{UL}/M_{all})]^2$$

$$[(Q_{UL}/Q_{all})^2 + (T_{UL}/T_{all} + M_{UL}/M_{all})^2] \cdot RF^2 + 2 \cdot (P_t/T_{all}) \cdot (T_{UL}/T_{all} + M_{UL}/M_{all}) \cdot RF + (P_t/T_{all})^2 - 1 = 0$$

... which is an equation of 2nd degree in RF with well-known solution:

$$A \cdot RF^2 + B \cdot RF + C = 0$$

$$RF = [-B \pm (B^2 - 4 \cdot A \cdot C)^{0.5}] / (2 \cdot A)$$

... as the independent C term is negative ($P_t < T_{all}$) and the RF must be a positive real number, finally:

$$RF = [-B + (B^2 - 4 \cdot A \cdot C)^{0.5}] / (2 \cdot A)$$

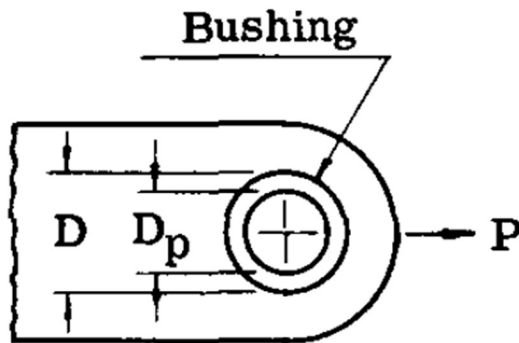
2.3 BUSHINGS

A typical check of bushings in pinned lugs configurations is to verify that the bearing forces does not cause permanent deformation in Limit Loads. As per Bruhn's D1.11...

Take A_{br} as the smaller of the bearing areas of bussing on pin or bushing on lug (the latter may be smaller as a result of external chamfer of the bushing, oil grooves, etc). The allowable yield bearing load on bushing is then:

$$P_{bush,bry} = 1.85 \cdot F_{cy} \cdot A_{br}$$

where F_{cy} is the compressive yield strength of the bushing material.



$$A_{br} = \min(D_p \cdot w_a, D \cdot w_b)$$

w_a : width/thickness in contact between bushing and pin

w_b : width/thickness in contact between bushing and lug

Figure 7. Calculation of "projected" bearing area of bushing (figure from Bruhn's Fig. D.1.11)

In addition, in configurations with (pre)tensioned bolts, where bushings are sustaining significant axial compression forces, it must be checked that the (ultimate) compression load does not cause failure of the bushing cross section, by comparing that force with the following allowable:

$$P_{bush,c} = F_{cy} \cdot A_{bush} = F_{cy} \cdot \pi \cdot (D^2 - D_p^2) / 4$$

2.4 BALL-BEARING

The pin and some of the lugs, typically the centre/male lug, shall be connected by means of a ball-bearing system. This ball-bearing system shall be normally a standard part (not an in-house design). Its strength shall be checked by comparing the transferred load with the allowable load defined by its technical specification.

ToDo

Pending definition of method for checking lug failure when out-plane loads are significant (pull out of ball-bearing from lug, root bending failure, ...).

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
1	-	Subscript designating relative to any of the clevis-lugs
2	-	Subscript designating relative to the single/male-lug
all	-	Allowable, at Ultimate Load level (subscript)
A	mm	Distance between most external faces of the clevis-lugs
A_b	mm ²	Cross sectional area of the bolt shank
A_{br}	mm ²	Bearing area (contact area between two parts transmitting compression forces)
b	mm	Lever arm for calculation of bending moment in pin-bolt
B	mm	Distance between internal faces of the clevis-lugs
D	mm	Outer diameter of pin, or lug hole diameter, from context
d_i	mm	Inner diameter in pin of hollow circle section
D_p	mm	Outer diameter of pin
e	%	Elongation, permanent deformation after tension failure
E	MPa	Young's modulus
F_{cy}	MPa	Yield compression stress
F_{tu}	MPa	Ultimate tensile stress
F_{ty}	MPa	Yield tensile stress
F_y	MPa	Yield stress (not specifying if in tension or compression)
g	mm	Gap between internal face of clevis-lug and nearest point of single/male-lug
I_b	mm ⁴	Bending moment of inertia of the cross section of the bolt shank
K_{sf}	-	Shape Factor, i.e. the ratio between the peak shear stress and the mean/averaged shear stress in the cross section
LL	-	Limit Load level (subscript)
M	Nmm	Bending moment in pin-bolt
P	N	Applied force in the single/male-lug
P_t	N	Pretension load in the pin-bolt
Q	N	Shear load in the pin-bolt
RF	-	Reserve Factor, i.e. ratio between allowable load and applied load
σ	MPa	Axial stress
t_1	mm	Thickness of the clevis-lugs
t_2	mm	Thickness/width of the ball-bearing installed between the pin and the male-lug
t'_2	mm	Thickness of the single/male-lug
T	N	Axial tension in the pin-bolt due to external loads

LUGS AND PINS

T_{all}	N	Tension allowable load of the pin-bolt
UL	-	Ultimate Load level (subscript)
w	mm	Width of bearing/contact area for bushings (typ. thickness of the lug)
y	-	Yield (subscript)

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 2	HSB 26101-01	Is. F	Static failure loads of metallic double shear lugs
Ref. 3	HSB 26101-02	Is. C	Static failure load of bolts in metallic double shear lugs
Ref. 4	[J. C. Ekvall's]	1986	STRENGTH ANALYSIS OF PIN-LOADED LUGS
Ref. 5	ESDU 91008	1991	STRENGTH OF LUGS
Ref. 6	HSB 52130-01	Is. D	Allowable bending moment of metallic circular tubes
Ref. 7	[W.C.Young, R.G.Budynas]	6 th	Roark's Formulas for Stress and Strain
Ref. 8	NT-T-ADS-95022	Ed. A	DISEÑO DE ELEMENTOS DE UNIÓN DE SUPERFICIES MOVILES
Ref. 9	[M. C. Y. Niu]	2 nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 10	[M.A.Melcon, F.M.Hoblit]	1953	Developments in the analysis of lugs and shear pins

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
STRENGTH2000 (Airbus DS)	4.2.1-rc1	Detailed sizing of metallic and composite aircraft structures (Modules HSB26101, HSB52130 and LUG)
FLEXA (Airbus DS)	3.5	Strength of beams sections in axial and bending loads

Table 2. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub- chapters affected
1.0 27/02/2023	G. Baños G. Moreda	First issue	All pages

Table 3. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Analysis (TEPAA)	-	F. Glock 27/02/2023
Stress Methods (TEPAP-TL4)	G. Baños 27/02/2023	-

Table 4. *Approval signatures table for last revision of this chapter*